Bounded End-to-End Capstone

Session 19 | 3 hours

One slice

Implement and review:

```
POST /api/bookmarks
```

Everything else is deferred.

The result is a handoff

- implementation-ready issue;
- patch or patch plan;
- executable test evidence;
- review decision;
- deferred scope and next action.

Keep the boundary in the issue

The accepted GitHub issue holds every scope decision. Implement only `POST /api/bookmarks`, use synthetic data, and keep the existing dependencies.

Stop if the work requires another endpoint, live data, deployment, tool configuration, or a new dependency.

Session 18 is an input

The accepted specification must separate:

Behavior	Implementation constraint
Observable request and response rules	Node.js, Express, in-memory storage
Validation and duplicate behavior	Existing dependencies only
Acceptance scenarios	Exact files and test command

Issue trace

Every acceptance criterion maps to:

```
specification ID -> planned file -> executable test
```

If the issue adds behavior missing from the specification, revise the issue before implementation.

Scope guard

In scope:

- create one bookmark;
- validate the accepted fields;
- reject a duplicate URL;
- return the documented envelope and status.

Out of scope: authentication, persistence, other endpoints, deployment, and new packages.

Path	Use
Approved agent	Implement the single issue
Manual	Implement the same files and tests
Blocked	Produce a reviewable patch plan

Executable verification

```
npm test
```

The suite checks success, defaults, field validation, and duplicate URLs.

Review and handover

Check the specification trace, diff, test output, dependency change, and scope decisions in the issue. Record any deviation in the handoff.

Record one decision: approve, request changes, or pause.

Lab time plan

Phase	Time
Accept the specification and scope	25 min
Write the issue and establish the baseline	25 min
Implement or write the patch plan	35 min
Verify, review, and hand off	35 min